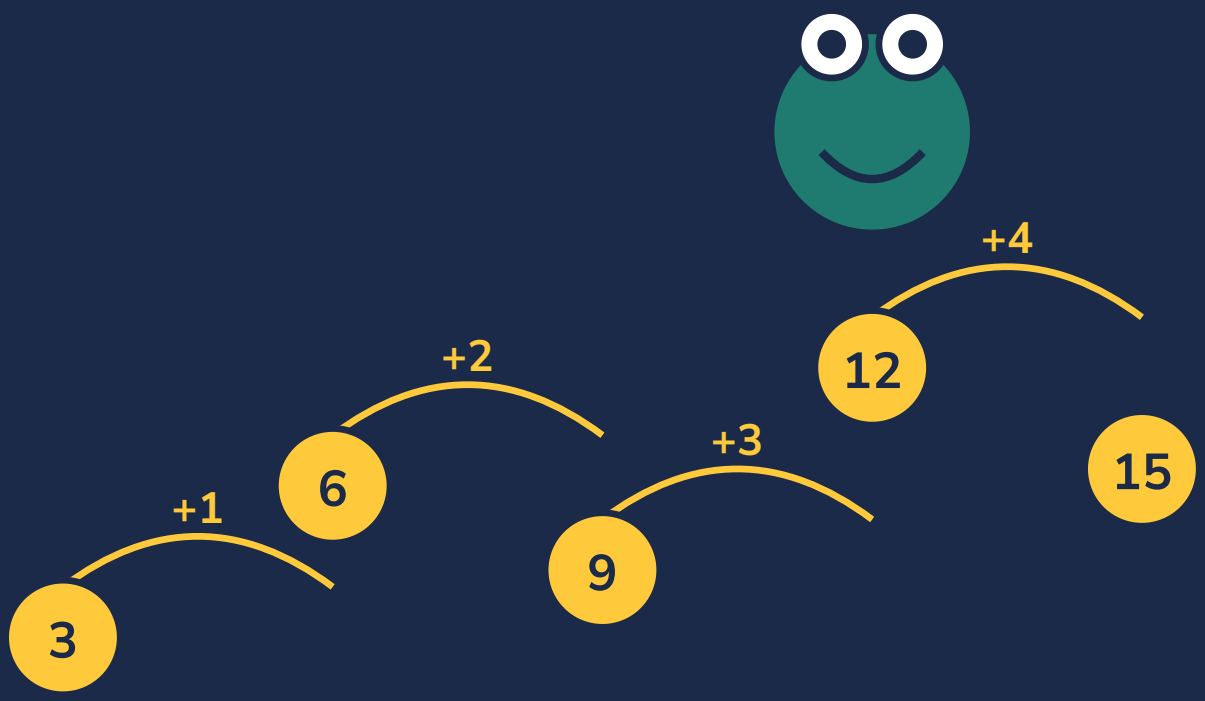
FREE TASTER • PRIMARY 1

Think Like a Mathematician

Six puzzles to try at home. About 10 minutes each. No shopping needed.



Notice it. Test it. Explain why.

From the Math Inquirers Holiday Program

Preciouslyours Pte Ltd

Four rules for every puzzle

1 Notice it
What repeats? What changes?

2 Test it
Try it with coins, buttons or spoons.

3 Check it
Does your answer make sense?

4 Explain why
Say how you know.

How to help (for grown-ups)

Ask before you tell. Start with "What do you notice?"

Use real things. Coins, buttons, spoons or pasta make numbers easy to see.

Stuck? Look again. Try it with objects. Try a smaller example. Say the answer last.

Ask "How do you know?" even when the answer is right.

Praise the thinking, not the cleverness: "I like how you checked that."

Keep it short. Stop while it is still fun.

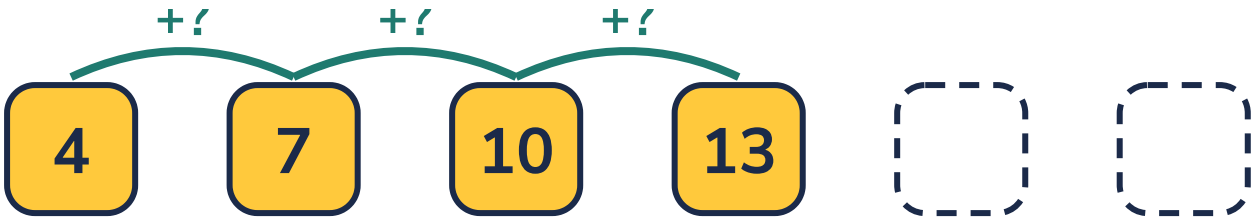
Hard question? Great! Mathematicians get stuck all the time. That is when thinking happens.

Puzzle set 1: Frog Jumps

Patterns

Find the rule, then write the next two numbers. Try it with a number line, or hop a coin along the table.

1 4, 7, 10, 13, ___, ___ What is the rule?



The rule is: _____

2 5, 10, 15, 20, ___, ___



The rule is: _____

3 1, 5, 9, 13, ___, ___



The rule is: _____

Think harder

Fill the gaps

4 3, 8, ___, 18, ___, 28

3	8		18		28

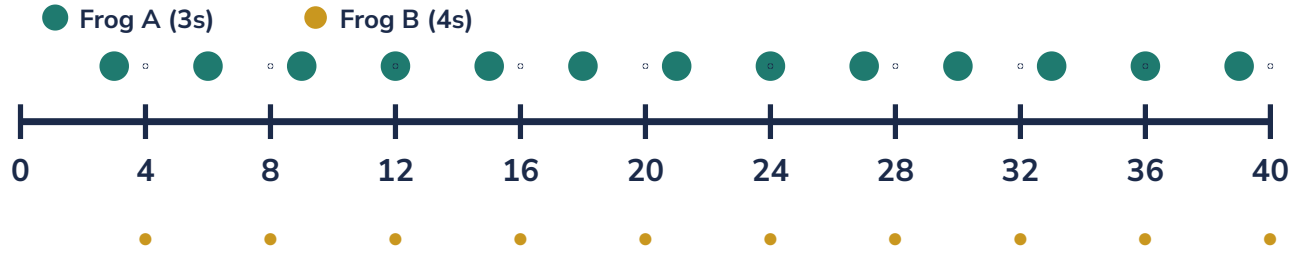
5 40, ___, 28, ___, 16, 10

40		28		16	10

★ Challenge ★

only if it is fun

6 **Frog A jumps in 3s from 0. Frog B jumps in 4s from 0. Which numbers up to 40 do both frogs land on? (Do not count 0.)**



Explain why: How do you know you found them all?

Try it with objects

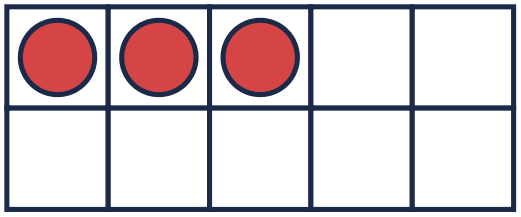
Line up 40 pasta pieces. Put a coin on every 3rd piece, and a button on every 4th. Where do the coin and the button sit together?

Puzzle set 2: Number Bonds

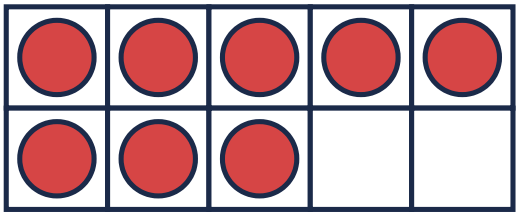
Making 10

Put 10 buttons or counters in two groups to help you.

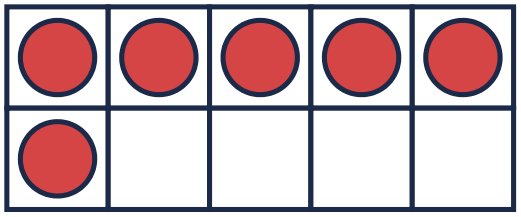
1 10 = 3 + ___



2 10 = ___ + 8



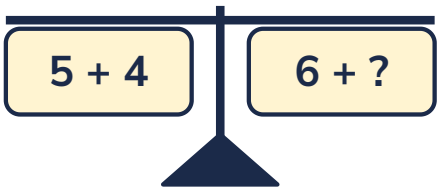
3 10 = 6 + ___



Think harder

Keep it balanced

4 5 + 4 = 6 + ___

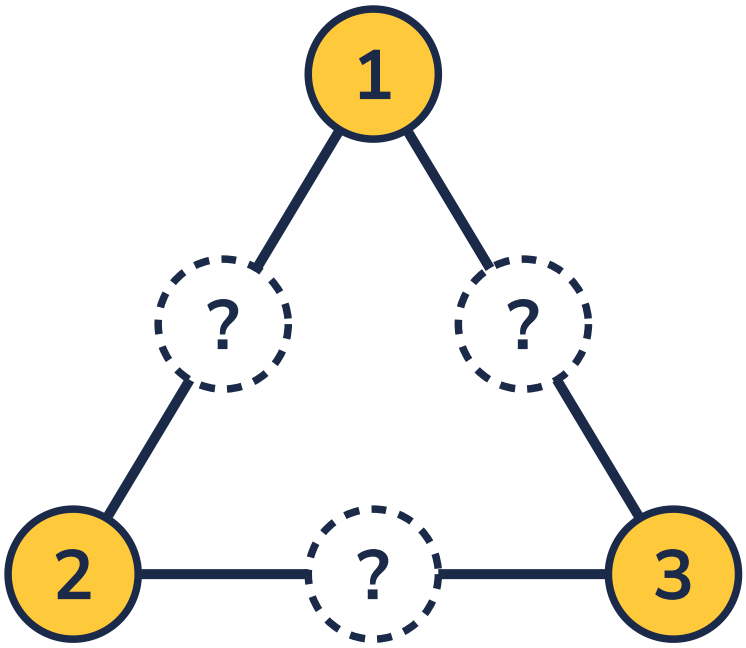


5 8 + 2 = 7 + ___

Both sides must be the same. Check by working out each side.

★ Challenge ★ — Magic triangle

6 **Put 1, 2, 3, 4, 5, 6 in the circles so that each side adds to 9. The corners are 1, 2 and 3. What goes in the middle of each side?**



Explain why: How do you know each side adds to 9?

Try it with objects

Write 1 to 6 on six bits of paper. Move them around until every side makes 9.

Answers

Look only after you have tried. A different way of thinking is fine, as long as you can explain it.

Set 1	Set 2
1. 16, 19. Add 3.	1. 7
2. 25, 30. Add 5.	2. 2
3. 17, 21. Add 4.	3. 4
4. 13 and 23. Add 5.	4. 3 (5 + 4 = 9, 6 + 3 = 9)
5. 34 and 22. Take away 6.	5. 3 (8 + 2 = 10, 7 + 3 = 10)
6. 12, 24 and 36. Both frogs land there because 12 is in the 3s and the 4s. Then it repeats every 12.	6. Between 1 and 2: 6. Between 2 and 3: 4. Between 3 and 1: 5. Each side adds to 9.

Want more?

Free weekly puzzles

Join our free list. Each week you get a new set of puzzles and a five-minute "explain why" question for the dinner table.

[Sign-up link or QR code]

Math Inquirers Holiday Program

Ten 90-minute lessons over five weeks, in small groups of up to 8 children. Face to face or online. Hands-on puzzles, a printed workbook with a cut-out kit, a stamped Inquiry Journal and a certificate.

[Dates, fee and sign-up link]

Questions? [Name, email or phone]

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